

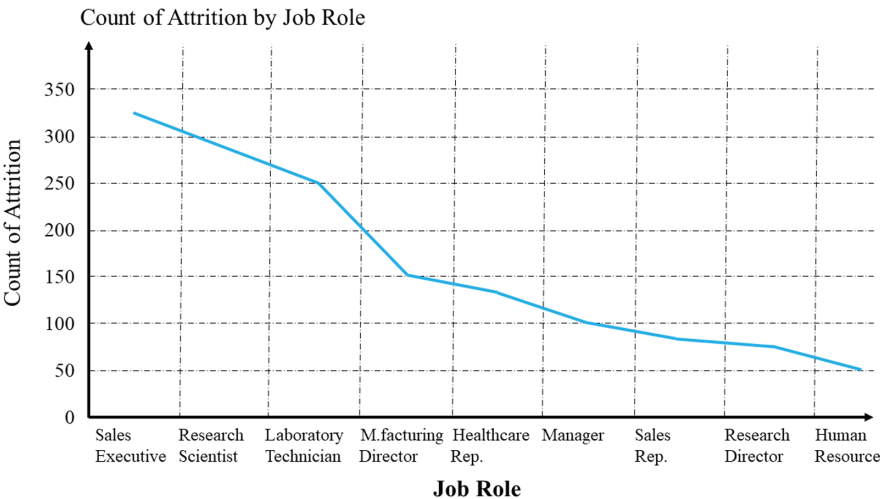


FIGURE 15.3 Attrition based on job role.

15.8 CONCLUSION

Even though organizations utilize workforce analytics to improve employee experience, they are not without challenges. Managing HR is critical to the organization. HR managers need tools to foresee some of the critical impact factors on one firm, such as turnover, future high performers, future skill requirements, cost of skilling, and corresponding benefits to the organization.

The analysis done in this chapter illustrates how the organization’s metrics greatly help identify the organizational problem. As a human resource, the organization should focus on addressing the critical issue to reduce attrition. To identify the organizational problem, the organization should focus on the critical metrics related to it. Once the key metrics are identified, they have to be analyzed for why the problem occurs.

With the dynamic and global environment, the measurements will be dynamic for identifying the problems of the human-based workforce will lead to much time. However, the problems will be fixed only by human resources, but identification or predicting the problem of the organization will be early using workforce analytics (Nigel et al., 2015).

According to Nigel et al., (2025), determining the best action and carrying it out can be the most challenging aspect of analytics. Analysis, measurement, and monitoring are only the beginning of analytics. Even if other teams carry out the interventions, it is crucial to help ensure that they are taken. This can involve enlisting the aid of other teams to demonstrate the value. It might take some time for the corporation to recover from those setbacks, and it would be challenging to keep up with the changing demands of the global business environment.

Finding new performance measurements is the only way to overcome these challenges. Metrics and analytics are thus widely used to assess the effectiveness of the company and support timely decision-making. The company should put plans in place to boost R&D, where HR specialists will experiment with new measurements and test presumptions about what the organization's current systems need. This experiment may inspire fresh ideas and enable the development of new measures and analytics.

Srivastava (2020) says that future generations are likely to view the creation of efficient workforce analytics and human resource metrics as a critical source of competitive advantage. Decision-makers will better manage and enhance HR programs and procedures by utilizing workforce analytics and HR metrics. This enhances the workforce's efficiency in turn.

By efficiently managing the company's human capital, management decision-makers can change comprehensive employment systems by utilizing metrics and analytics in all HR department functions.

Levenson (2017) states that HR managers must demonstrate their ability to use metrics and analytics to run their operations so that others will be more receptive to their advice. To improve their potential to produce the desired results, operational departments' business partners and HR specialists must collaborate closely.

Any information system that generates metrics and analytics will lead to better decision-making by encouraging decision-makers to choose new, better options. Finally, applying workforce analytics and HR metrics enhances the organization and managers' ability to make strategic decisions that benefit the business's overall performance.

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16 Robotic Process Automation (RPA) Applications and Tools for the Workforce Management System

Sathya V., Jayashree K., and Malathi S.

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16.1 INTRODUCTION

Various tools and approaches are included into robotic process automation (RPA) with the goal of automating repetitive and structured service activities that were previously carried out by people (Tailor, Pareek & Khang, 2022). This is accomplished by using software algorithms, sometimes referred to as software robots or bots, which imitate the front-end execution flow of people.

Robots can log into multiple applications to extract, process, and enter structured or semi-structured data from various sources, just like a human user would, and they can interact with the user interface through mouse clicks, keyboard interactions, and the interpretation of text and graphics.

Because RPA sits on top of information systems and only accesses applications through the presentation layer, back-end systems typically do not need to have defined interfaces.

Therefore, the robots carry out tasks without interfering with human activities and do not require application programming interfaces (API) to transfer and process data. Here are four advantages of RPA.

- RPA is simple to set up on different computer operation systems.
- RPA interfaces operate by simply dragging and dropping process stages. A code is produced on the back end automatically when users drag and drop icons to automate a procedure.
- RPA software sits on top of current systems without requiring the creation, replacement, or future development of pricey platforms. It uses a human-like interface to access other computer systems (through the user interface). No program is impacted by RPA, since it uses the presentation layer to access other systems.
- RPA is secure for businesses. RPA is a strong platform designed to satisfy business security, scalability, and auditability needs. Robots can be scheduled in a job that runs on multiple robots using the RPA orchestrator. For centralized and efficient processing, the RPA orchestrator also offers robot auditing and monitoring.

This chapter contains six sections.

- The chapter’s introduction for RPA technology is shown in [Section 16.1](#).
- The chapter’s related work for RPA is shown in [Section 16.2](#).
- The technology and applicability of RPA are covered in [Section 16.3](#).
- The implementation procedure of RPA is discussed in [Section 16.4](#).
- In [Section 16.5](#), the usage, scalability, advantages, and disadvantages of RPA for various business organizations are explained.
- The conclusions and future scope for RPA are defined in [Section 16.6](#) of our chapter.

16.2 RELATED WORK

The old hand-based audit approach is being disrupted by RPA, as this procedure involves a lot of repeated tasks and physical work. Many studies also discussed the benefits of RPA in auditing, since it allows software bots to handle low-level, routine tasks and make midlevel choices, improving the organization's productivity while maintaining its organizational structure.

Additionally, they emphasize how it will alter the function of auditors by eliminating menial tasks and limiting them to higher-order-thinking tasks, which would ultimately improve the caliber of work done in the organization. RPA's distinctive features, which set it apart from other automation paradigms, are found in business process re-engineering.

The areas of risk in RPA that are connected to strategy, sourcing, tool selection, project time estimates, operations and execution, and change management are also identified (Hajimahmud et al., 2022).

The effectiveness of RPA on front- and back-office occupations is discussed by authors Tailor et al. (2022). Jobs in the front office include selling, customer service, and managing needs, whereas those in the back office include finance and human resources.

RPA is best suited for high-volume, standardized operations that are particularly rule-driven and do not need intellectual judgement. RPA works well for back-office tasks, including billing, travel, expenditures, accounts payable, and accounts receivable.

Cost savings and mistake reduction have resulted from the use of RPA to back-office procedures. The impact of RPA on global accounting services is further discussed by authors Tailor et al. (2022).

RPA, also known as intelligent robotic process automation (IRPA), is employed in the finance and accounting (F&A) industry. The study demonstrates that implementing RPA has a major impact on people and businesses by reducing work and personnel numbers.

The long-term benefits of RPA outweigh any short-term economic and employment disturbance, since it generates new job possibilities and alters the nature of current ones.

16.3 RPA TECHNOLOGY

Recent years have seen the effective use of artificial intelligence (AI) algorithms (Rana et al., 2021) and machine learning (ML) techniques in real-world contexts, including business, industry, and digital services.

AI algorithms (or techniques) that often reflect the paradigms/approaches of rational characteristics such as connectionism, genetics, statistics, and probabilities, based on cases can be applied ML (Cewe et al., 2018), which we can use to “teach” machines how to deal with data more efficiently by simulating the learning concept of rational beings. In this way, it can help us to examine and extract information using AI algorithms and an ML methodology in order to categorize, associate, optimize, group, forecast, and detect trends more exactly.

Given the range of applications for AI, RPA has steadily included implementations of algorithms or AI approaches used in specific settings (for example, enterprise

resource planning, accounting, and human resources) to classify, identify, and categorize its automation features.

16.3.1 RPA TOOLS

Tools for RPA are frequently used to set up job automation. For the automation of repetitious back-office tasks, these tools are essential. There are several RPA tools on the market, so picking one might be difficult.

16.3.1.1 Automagica Tool

The source code for the proprietary Automagica tool Forrester (2014), which is accessible on GitHub, is open-source and intended for non-commercial use. It was primarily created using the Python programming language; and however, the community may use and customize for the different implementations (e.g., AI techniques or algorithms).

RPA includes a number of fundamental features, including the ability to read optical character recognition (OCR), extract text from PDF files, automate information in Word and Excel documents, gather data from the web, and create automation processes. It also enables integration with Google Tensorflow for text and image recognition.

16.3.1.2 ZAPTSET Software

Any organization that uses software for operations or development and is looking to automate its back-office operations or software testing processes, as well as develop a seamless and iterative automation framework, should consider ZAPTEST as a software automation solution.

The functionalities of ZAPTEST tools are automated functional and performance testing, RPA, execution of the cross-platform (parallel) test plus RPA software automation in a seamless manner, DevOps (mock-ups) API testing with automation, and auto-documentation.

16.3.1.3 Help System

Help Systems' Automate is a complete RPA platform created to automate manual, time-consuming tasks within your business. Due to its simplicity of use and capacity to expand your automation plan economically, Automate is designed and priced appropriately for employee-driven, enterprise-wide automation. Additionally, the no-code automation gets you up and running quickly, so you can see an instant ROI for both GUI and back-end operations.

The functionalities and features of Help System can be given as over 70 native actions with form-based development and drag-and-drop, to combine tasks, apply criteria, and implement flow control to construct logical workflows, use a graphical workflow designer, step recorder quickly and easily creates UI-driven automation with the option to incorporate business logic for more robust records.

Help System also has a powerful built-in API integration kit to connect to the programs that operate your company to better manage unstructured data. Nowadays, AI and machine learning (Khanh & Khang, 2021) provide intelligence to business activities with consumer mindset analytics, so the automate bot store, an online

marketplace, or online services want to boost the return on investment (ROI) and use free pre-built bots and virtual assistants (Bhambri et al., 2022).

16.3.1.4 KOFAX Tool

One of the most effective ways to gather, improve, and transmit data from any application or data source is using KOFAX RPA. Websites, portals, desktop programs, and business systems without any coding are all included in this.

The noted features of KOFAX include elimination of mistakes by automating data collection and input; boosting efficiency of operations with intelligent software robots; process monitoring and improvement with robotic process intelligence; robot deployment, management, and operation from a centralized server; analytics; and built-in process intelligence.

16.3.1.5 Power Automate Tool

Power Automate is an RPA tool that can help you increase productivity within your business. You can create a secure workflow using this application without writing any code. The tool enables us to improve worker precision and effectiveness. This software offers a step-by-step manual to make business process automation simpler.

MS security technology is used to protect Power Automate tool automation. It provides AI models for text recognition, object detection, and form processing. This application is compatible with Microsoft 365, Azure, and Dynamic 365 (Khang et al., 2022a).

16.4 DEPLOYMENT PRACTICES IN RPA

Any company is using RPA, a cutting-edge technology, in its system. Once RPA has been deployed, a company is no longer dependent on other businesses or service providers to carry out a certain set of duties on their behalf. RPA deployment is broken down into four stages, which aids the business in creating a concise business case for the organization (Khang et al., 2022c).

16.4.1 FIRST STAGE: RECOGNIZING THE POSSIBILITIES OF RPA

During this stage, the business undertakes internal research and workshops to understand its potential and pinpoint the biggest issues and challenges it will face. The business assesses the effectiveness of its present procedures throughout this phase. They are able to pinpoint the regions where RPA can be used by doing this.

Understanding the organization's business issue is crucial for effective RPA deployment. Only then should search for potential solutions and the appropriate RPA software application to automate the company begin. This stage might last for a day or two.

16.4.2 SECOND STAGE: PROCESS EVALUATION

The most important step is determining how accurately the employees are carrying out the organization's primary duties. This aids the business in determining whether using RPA would be advantageous for it. There could be a list of procedures that can be automated after the assessment. Then this list is ranked in order of possibility.

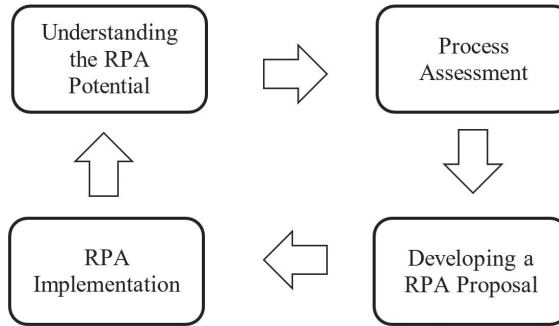


FIGURE 16.1 Stages in RPA deployment.

Process evaluation must be carried out by an RPA expert; otherwise, procedures that cannot be automated or do not require automation would be added to the list, wasting valuable time.

RPA has limits, and it is crucial to be aware of them. The RPA program won't be able to automate a process if it is too complex. Consequently, the business expert must be able to perceive the potential as a whole.

The RPA program is designed in a way that allows users to create products that will be useful in the future. For instance, if the person is unsure if a specific activity needs to be automated, they can create a foundation so that it can be examined later, avoiding the need to start from scratch, as shown in [Figure 16.1](#).

16.4.3 THIRD STAGE: CREATING A PROPOSAL

Provides all the business cases and relevant information that is necessary to understand if the job is justified and independent of the necessary investment. Because a business requirement and a benefit analysis showing how much money may be saved are two items that frequently appear in business cases. So the business proposal is created using the data acquired in step 2 in its entirety.

The business plan would include the company's strategy for automating the procedures and how the various systems could cooperate to maximize customer happiness, cut costs, and boost productivity. It is preferable if the business provides a functional model or prototype of how the system would function when RPA is implemented.

In the eyes of the organization, this aids the business in building trust. Additionally, the proposal must outline the costs the company will incur and the time needed to build and install the program.

16.4.4 FOURTH STAGE: RPA IMPLEMENTATION

The firm may proceed with developing an operational RPA program if the organization accepts the plan it has provided. The company's specialists create a step-by-step

manual for the robot to follow in order to do the particular set of duties. We refer to this step-by-step manual as an internal library.

16.5 ROBOTIC PROCESS AUTOMATION AND BUSINESS PROCESS MANAGEMENT

Investigating the parallels, differences, and complementarities between RPA and related technologies is crucial, as stated previously. In this regard, Ovum (2015) urges the business process management (BPM) research community to look into business process management systems (BPMSs) and RPA integration, because RPA and BPM are adjacent fields with complimentary aims.

BPM is a multifaceted strategy that aims to improve corporate performance through process optimization, continuous process improvement, and digital transformation. One of the BPM initiative's unavoidable viewpoints is BPMS as a comprehensive software platform that includes a wide variety of features including process design, analytics, and monitoring (Mendling et al., 2018). RPA, on the other hand, handles discrete, repetitive jobs and runs processes much like a human would.

Cewe et al. (2018) state that "RPA is responsible for repeated sequences of activities that may be totally assigned to software robots," whereas BPMS is used to coordinate end-to-end processes and handle interactions between people, robots, and systems.

The authors from business practice (Mendling et al., 2018) strongly recommend combining both of these technologies even though they are frequently used separately to increase the business value.

16.5.1 SOLICITATIONS AREA OF RPA

RPA is used often by global corporations for daily operations. These businesses use RPA for their normal procedures because it offers precise, consistent results with high production rates at comparatively low costs. [Figure 16.2](#) displays the top RPA applications across several industries.

16.5.2 FINANCE INDUSTRY

In the finance industry (FI), bots are most frequently used to duplicate financial data (such as client information, market prices, and credit ratings) across various internal systems and make it accessible to business units (Rani et al., 2023).

FIs started using RPA bots for this reason as early as a research by E&Y (2017) as shown in [Figure 16.2](#). The following are some prospective RPA application areas in finance:

- The process of reconciling the bank accounts.
- Financial and external reporting.
- Sales ordering and invoicing.
- Inventory management.
- Receivables and payables management.
- Fixed asset management.

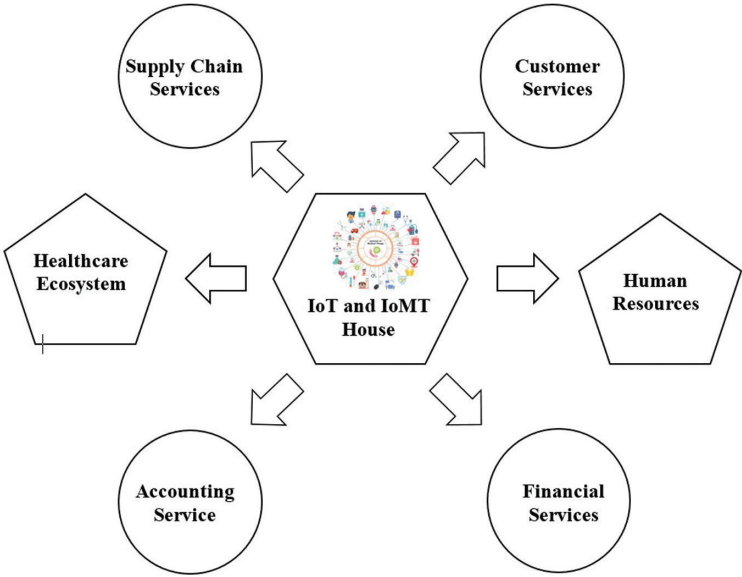


FIGURE 16.2 Application of RPA applications.

These uses of RPA are included on this list, which is consistent with lists from previous research. The most often mentioned application areas are accounts payable, reporting, and reconciliation. The heat map below displays the frequency of occurrence of these applications across various financial application subdomains (Khang et al., 2022c).

The financial applications where RPA has been utilized most frequently are indicated in blue-gray in these sections in [Figure 16.3](#).

The secondary goals for RPA applications in the banking industry are highlighted in gray in [Figure 16.4](#).

Another typical application area for the RPA bots in FIs is monitoring and screening. Bots have been used to continuously monitor application systems and networks, and they produce alarms based on supervised or learned heuristics. These notifications are also adapted to the demands of FI in order to escalate possible problems in accordance with the specified risk tolerance.

Additionally, bots are set up to escalate problems to human operators in the event of unusual conduct. In this case, robots and people may cooperate, passing tasks back and forth as necessary. Robots record the decisions and actions of the operator, then continue automation until the next decision-making stage.

16.5.3 BENEFITS OF RPA

There are two types of RPA benefits at a corporation: primary and secondary. The primary advantages are those that are directly connected to the application of RPA. Improvements in productivity, accuracy, and personnel reduction are a few examples.

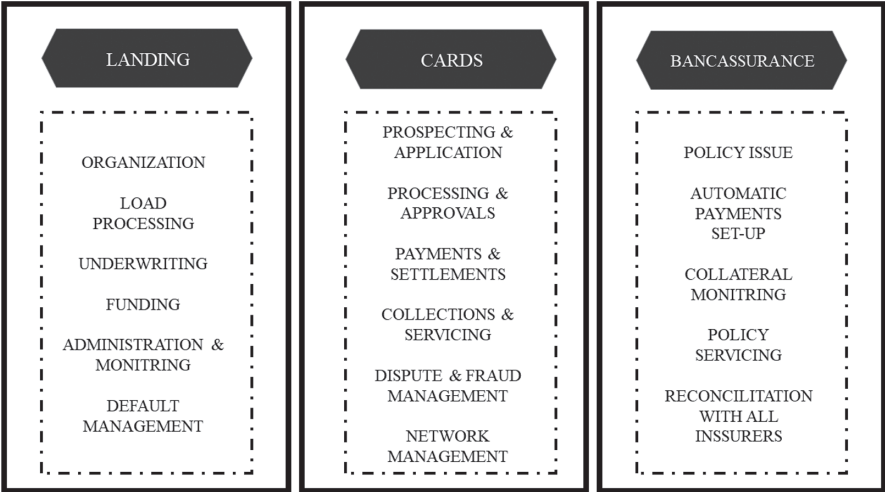


FIGURE 16.3 RPA heat map for various financial applications.

Secondary advantages are those that the implementation of RPA indirectly makes feasible. Even though they are more difficult to measure, these advantages can have a significant positive impact on the businesses’ bottom-line cost reductions as well as top-line revenue growth.

16.5.3.1 Increased Efficiency

Possibly the most frequently mentioned (and most readily understood) advantage of automation, increased efficiency is attained through increased throughput rate and

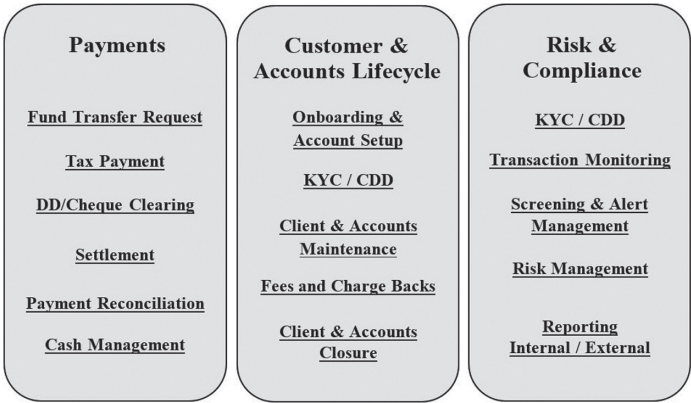


FIGURE 16.4 RPA heat map for the banking industry.

workload capacity as well as increased speed. This might enable a bank's loan closing and booking team to handle more loans each day, enhancing customer satisfaction and obviating the need for hiring as business grows.

16.5.3.2 Greater Accuracy

Any procedure that calls for human data entry or data translation increases the risk of inaccuracy. Humans become tired and distracted easily, but automated processes can work continuously, year-round, using computer memory and programmable rules to guarantee correctness and consistency. Data entry and transcription mistakes at a bank can lead to more work being redone, decisions being made based on false information, negative financial effects, and fines.

16.5.3.3 FTE Reduction

While it may be the hardest to convey, head count reduction is also one of the most misunderstood benefits. Redistributing the workload and reducing the need for future hires are more practical solutions. By embracing automation, a business may meet increased capacity brought on by expansion without increasing its personnel.

16.5.3.4 Workload Redistributions

This allows a company to transfer personnel from one area to another to meet organizational demands. This process is known as asset redeployment. If automation increases capacity but the workload doesn't keep pace, the extra labor supply might be distributed to different sectors. Think about how a bank would reorganize its front, middle, and back offices if it were suddenly given extra "free" time to work with.

It's important to remember that the terms "resources" and "capital" may apply to both human capital as well as technological resources like computers and bandwidth. All of the major benefits previously stated have to do with cost reductions, but when seen from a different perspective, these primary benefits improve the customer experience, which may also result in higher revenues.

For instance, if employees' capacity grows (via speed, efficiency, and workload redistribution), more time may be dedicated to tasks and activities that include interacting with customers (like improved customer service, increased sales efforts, and customer retention activities). Increasing speed to market through quicker loan decision, and approval or improving competitive positioning through cheaper costs (reflected in decreased lending costs and interest rate margins owing to resource needs and efficiency improvements) have advantages as well.

16.5.3.5 Faster Scalability

In the current banking climate, senior management should think about how to instantly scale their bank without adding more employees. This topic rises to the forefront when financial organizations expand both organically (additional sites or branches) and through acquisition. An automated workforce is dispersed as a software resource, whereas physical recruiting takes time.

You get a digital workforce that is already trained when you automate. Because additional jobs may be added by acquiring software licenses and implementing them

into an already existing automation protocol, scalability is immediate. Simply hit “go” after loading an automation workflow into a robot.

16.5.3.6 Increased Job Satisfaction

According to frequent job satisfaction surveys, the most engaged employees believe they are carrying out important work that is directly related to corporate performance measures. There are actions banks can take to improve job satisfaction and staff engagement. Start by eliminating automatable tasks so that these personnel can focus more on the needs of the consumer. If banks are not automated, they should think about who will perform this function in the future.

Although cost savings are frequently highlighted as the biggest advantage for banks deploying RPA, it's critical to go beyond ROI and delve further into RPA's capabilities to properly appreciate the possible benefits that come with using this technology.

16.5.4 ADVANTAGES

The software and services for automating robotic processes are prepared to run programs in the same way that a human operator would. With the use of rules, intricate tasks are handled automatically by the work flow.

RPA offers a wide range of benefits, including:

- **Ongoing service:** Once operating a true 24/7 service is involved, software package robots become clear in what they are performing and do not require breaks.
- **Scalability:** Processes that are set up for one software robot are frequently expanded to any number of additional robots, and vice versa; robots are frequently retired from one process to work on another.
- **Sincerity:** Once given a duty, robots are programmed to follow the directions exactly and without deviation.
- **Time:** Using robots, typical undertakings may be completed in a matter of weeks as opposed to the years it takes to do so with people.
- **Increased Productivity:** The miracle of RPA is that it's created to free human workers from their everyday repetitious jobs. As a result, these duties and operations are handled by technology, they operate significantly more quickly and then perform better.
- **Increased Productivity:** When technology handles the labor-intensive tasks, like RPA does, production may rise significantly. Additionally, knowledge workers will have more time to devote to important initiatives that promote innovation and development.
- **The Abolition of Human Error:** Even the most meticulous human worker may occasionally make an error. Unfortunately, these mistakes frequently end up being quite expensive. This danger is minimized by robotic process automation technologies, increasing accuracy (Rani et al., 2022).
- **Cost Savings:** Although implementing robotic process automation does involve an initial investment, the total gain in productivity and efficiency as well as the decrease in human mistakes more than justify the cost.

- **Lower Turnover:** Once employees are no longer hindered by monotonous, repetitive, and menial jobs, levels of satisfaction may naturally increase. Additionally, employees might value the opportunity to take part in more high-level initiatives, which contributes to greater worker satisfaction and retention.

16.5.5 DISADVANTAGES

- **Financial Costs:** Budgetary restraints are among the top reasons why firms choose not to deploy RPA. **Lack of technical proficiency:** Many of us think that the end user has to have substantial technical proficiency in order to benefit from robotic process automation. Sometimes, this belief prevents individuals from taking use of the many benefits that are available to them (Vrushank, Khang & Rani, 2023).
- **Major Change:** Doing a replacement technology needs to be modified, but with the right tool, the impact of that change is considerably less observable and disruptive than many expect. **Redundancy:** People who oppose RPA frequently worry that robots will take the jobs of people, despite the fact that the technology's primary goal is to assist people in their work (Hussain, Sivakumar & Khang, 2022).

16.5.6 ANALYTICS AND REPORTING

Pre-populating complicated periodic reporting needs for scorecards and dashboards (Khang et al., 2022b) are as follows:

- Data capture and cleaning to facilitate automated report production
- Produce scorecards and dashboards
- Gather information from many sources
- Archive information for multi-period analysis
- Automate the development and approval procedures for purchase orders
- Optimize the buying channels
- Data management for procurement
- Bill of materials updates
- Contract data management
- Vendor master data management
- Material master data management
- Automating the classification and processing of spending categories to determine strategy and track performance
- Contract management and the development of automated reporting and workbooks as important strategic inputs
- Strategic sourcing includes: Automating contract formation, modifications, and administration
- Ongoing contract compliance and monitoring
- Analysis and market research automation for suppliers
- Determine and choose providers based on predetermined standards

- Supplier qualification and setup
- Reverse auction automation
- Supplier relationship management
- Updates to the supplier site and connection with internal and external systems
- Advanced shipping notifications
- Integration of other important performance management data through automation (on-time delivery, quality, cost, etc.)
- Risk management for suppliers
- Monitoring and real-time alerting of risk occurrences
- Complex/outsourcing contract compliance management
- Inbound product quality testing and compliance reporting
- Consistent operational risk administration (e.g., financial, legal, reputational, etc.)

16.6 CONCLUSION

By implementing scripts, macros, and algorithms into the process, RPA uses both software and strategic techniques to successfully automate the calculation and design processes. RPA automates workflow, back-office operations, and infrastructure in a business, replacing workers who do repetitive jobs like clerical labor with machines (Khang et al., 2024).

By using virtual workers instead of real employees, they transform the workplace atmosphere and guarantee 100% error-free productivity, as opposed to people who can only demonstrate an average output of 60%. The regular procedure may be changed with just a few lines of software code. RPA technology is a perfect replacement for outsourcing different duties.

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17 Exploring the Concept of Managing Women Employees' Work-Life Balance in Information Technology Company

Jansi Rani T. and Krishan Priya V.

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17.1 INTRODUCTION

In the olden days, women played multiple roles at home because they had to take care of their spouse, children, elders, and other family members. The male's attitude was that the women had to take control of the family and all the needs of the home, but

the situation has changed, and women now have started going out for their professional career growth and have formed a strong identification (Khang et al., 2023a).

In number, women have increased towards various categories of opportunities for professional work. Working women have gained interest in this dynamic environment because they work. In each one, there will be distinct demands in the profession and they have to handle them without losing their expectations for personal life (Babasaheb et al., 2023a).

In a woman's life, balancing a personal and professional life are two sides of a coin. In the IT industry, the working women have to face the challenges of both personal and professional life (Khang et al., 2023b).

When working in the IT sector, it is critical for employees to meet their daily and monthly target and achieve the desired outcome. As a result, professionals in IT have a rigorous schedule, and it is always a challenge to strike a healthy balance between their professional and personal lives. As a result, managing women's professional and personal work is a difficult challenge (Veluthan et al., 2020).

By concentrating on the personal and the professional work, the employees feel happy as well as suffer about their work-life balance, that's the thing is dealt in this chapter. The IT field is one of the main fields in outsourcing. It is also gaining economic status in society. The IT industry operates 24/7. Women have to work the various shifts allocated to them and also have to manage their personal responsibilities (Snehal et al., 2023).

Work-life balance (WLB) is a big concern for women employees in the current environment, as prolonged work hours in the workplace detract from women employees' productivity and time spent with their families (Vasumathi, 2018). The requirement for autonomy and flexibility in dealing with stress factors and WLB is critical for female employees (Chakraborty et al., 2020).

The work pressure of employees has altered at work as a result of major improvements in technology and increased worldwide competitiveness. In the present day, the question of work-life quality has received a lot of attention from the employers' perspective (Babasaheb et al., 2023a).

Organizations must focus their attention on fulfilling a balance between work life and personal life among employees in order to keep talented and skilled employees (Swarochi et al., 2018).

17.2 CONCEPTUAL MODEL

For everyone in the world, the first and foremost thing in life is family. Family is one of the most crucial social systems that all universal aspects want to preserve, because it is the foundation of society. Here, for an IT woman employee who is balancing personal and professional challenges, she definitely should take care of members in the family (Babasaheb et al., 2023b).

Whenever the situation is needed, women have to play the role of mother, spouse, and daughter-in-law. Involvement with family members is that of spending time with their family; educating their children; supporting the spouse by discussing day-to-day activities, the future, and family decisions; and taking care of elders in the family.

Family is a social group in society made up of individuals with ties to one another through various relationships. Culture is a particular collection of values, norms, and

traditions from a particular period and location. When combining these terms, we get the meaning of family culture as a particular set of values, norms, and traditions adhered to by a social community of connected individuals. Nowadays everyone knows that the IT is one industry which has various rotational shifts, including with night shifts (Snehal et al., 2023).

People have the awareness of sending their children to school and fetch them after work, especially women to the night shift job, practically it’s also one of the family culture changed now. The IT women employees keep in mind that she has to adopt the family culture which is followed in their family as well as to manage with this work-life balance.

As mentioned above, under the personal challenges, the woman employee has to be involved with the family, which means she has to spend time with her spouse, children, and other members of the family, and also supporting the family culture while still getting work done at the company (Babasaheb et al., 2023a).

Long and irregular working hours are widespread in all types of industries. In the IT industry, working for longer time during the day leads to elevated stress levels, unhealthy eating patterns, inactivity, and disease. This is one of the main professional challenges a woman employee has to contend with to balance life.

With more working hours, the IT industry will have more job responsibilities to finish the project within turnaround time (TAT), i.e., the period of time it takes to execute a job or respond to a request from the client side (Gupta et al., 2023).

Women employees, whatever personal issues they have, have to work more hours to complete the given job. It’s one of the main responsibilities to maintain the job. IT women employees have to face both personal and professional challenges when managing the work-life balance (Vrushank et al., 2023a).

Employees benefit from training and development programs, resources, mentoring, and demanding job opportunities provided by IT businesses’ workplace culture and environment.

Women who are courageous and strong enough to overcome these challenges serve as role models for others, showing that everything is possible for those who truly want it (Pareek et al., 2017) as [Figure 17.1](#) shows.

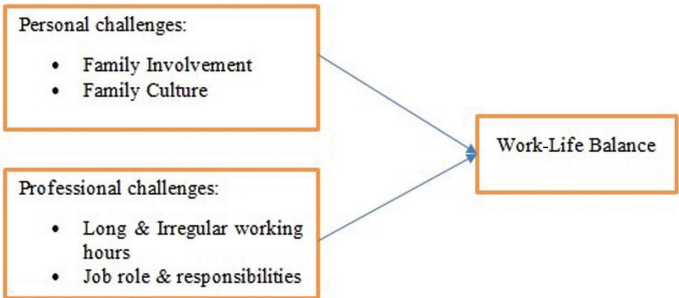


FIGURE 17.1 Work-life balance of IT women employees.

17.3 RELATED WORK

To examine the various personal and professional challenges of IT women employees. To analyze the personal and professional challenges towards the WLB of women employees in an information technology field (Khang et al., 2023b).

17.4 REVIEW OF LITERATURE

17.4.1 FAMILY INVOLVEMENT

Witzel et al. (2022), researched whether gender and family participation affected daily variations related to interpersonal stressors. According to multilevel model findings, relationships between daily interpersonal stressors and their effects were not significantly moderated by family participation. Based on the findings, interpersonal stress on a daily basis may be bad for emotional health, regardless of whether there is family involvement. Gragnano et al. (2020) mainly researched the family domains and work.

The impact of work-health balance and work-family balance on job satisfaction was explored. Workers are aware of the relevance of the health domain in creating a good work-life balance, according to this survey.

Rangarajan (2018) identified the working women's work-life balance. The author stated that working women face several problems both in the office and in their home. They also suffer mental and physical health problems.

Finally, concluded that achievement and enjoyment will contribute to the balance of work-life in all fields. Tariq et al. (2012) measured that work-life balance is priority for both organizations and their employees, especially in today's vivid organizational scenarios.

The win-win situational tools for employers and employees are also work-life balance policies. Employees and priorities of an organization are successfully achieved and met along with the employees' personal needs.

17.4.2 FAMILY CULTURE

Dehkonov (2020) researched the implications of modernist perspectives on the question of family culture in Islamic religion in practical terms. Young families' life processes show the modernist characteristics of the culture of family relationships, which has incorporated religious charters. The researcher concluded that young, newly formed families with morally and spiritually mature members will have a significant impact on other families and society as a whole if they are able to communicate with one another.

Awasthy et al. (2021) articulated more about the worldview of aspiring Indian management professionals. The research proposes various layers in Indian culture, including basic assumptions, values, and artefacts.

Bulger et al. (2007) investigated the impact of the Indian cultural setting on the knowledge management process. The knowledge management process is influenced by several aspects of Indian culture; however, this effect is complicated due to conflicting cultural traits, and their combined effect may make comprehending the knowledge management process in the Indian cultural context difficult.

Brown et al. (2019) articulated about the women's feelings toward the influence of societal and organizational culture on their usage of work-life balance programs in the two countries. This study made a significant contribution by examining three factors linked to the use of work-life balance programs: societal culture, organizational culture, and work/non-work self-efficacy (an individual-level component).

Baral et al. (2010) examined the role of work-family enrichment in relationships with organizational measures of job features, work-life benefits, policies, superior support, work-family culture, and job performance for work-life balance.

17.4.3 LONG AND IRREGULAR WORKING HOURS

Voglino et al. (2022) articulated about the review to evaluate the connections between reduced work hours and health outcomes. According to research, reducing working hours while maintaining pay may be a useful workplace intervention to improve employees' well-being, particularly in terms of stress and sleep.

Hsu et al. (2019) examined the relationships between hours at work, job satisfaction, and work-life balance. Furthermore, they explored whatever observed time management helps to moderate the impact of hours at work and work stress. The authors suggested that labor laws have been changed as necessary to avoid excessive hours of work and strengthen flexibility at work time.

Banu et al. (2015) articulated the work-life balance initiatives under four categories: flexible work options, leave benefits, dependent care benefits, and employee support schemes and services. These were ranked based on the presence and absence of importance.

Yu (2014) articulated that flexible working hours-based work-life balance policies are just a partial compound, and succeeding improvements in work-life balance look to recognize higher occupation outcomes safety and decreasing job power, both in work areas and in discussions of policy.

Sudhakar et al. (2010) investigate the inhibiting factors that obstruct the work-life balance. simultaneously, they seek to describe work-life options preferred by employees. The analysis was conducted, which belongs to the organization of IT and IT enabled services might adopted to boost the balance of work & life. The work-life balance choices listed with his research were family/friends program, flexible working hours, professional counseling, career guidance and life skill program.

Sturges et al. (2004) investigated the causes, the condition of psychological contract status, possible implications, and, in particular, how work/non-work conflict is related to hours worked and organizational commitment. This study explored the degree to which the college graduates value and achieve WLB successfully and the impact of work and non-work conflict on graduates' organizational duty.

17.4.4 JOB ROLE AND RESPONSIBILITIES

Gordon (2022) researched how the restaurant managers are motivated to work in a job known for long hours and high intensity. These managers had a lot of responsibility, worked more than 48 hours a week in chaotic, fast-paced situations, and found it challenging to take time off. The fast-paced nature of their work should

not diminish, according to managers, but rather should become less physically and emotionally taxing.

Cavlan et al. (2021) demonstrated the coping mechanisms of self-employed women who work in nursing homes in Northern Cyprus in terms of work-life balance. By focusing on the contextual variables, the research contributes to the work-family literature by highlighting the various methods women used to be able to achieve work-life harmony in nursing homes.

Wheeler et al. (2020) researched the self-management and work performance of individuals. Individuals who have a better understanding of themselves and how they affect work performance are better able to carry out their responsibilities and meet their work needs. Based on the findings, the research implies that exercising self-management on a daily basis can help a person become more self-aware, which can improve a person's work performance.

Gupta and Singh (2018) investigated the relationship between distributive, procedural, and interactional justice and the intention of IT experts to stay with the company. The authors suggested that employee retention is influenced more by perceived fairness in interpersonal treatment than by the distribution of organizational rewards and the fairness of organizational procedures.

Chan et al. (2017) investigated the mediating features of family conditions, job, and work-life balance with the interaction linking self-efficacy and engagement at work. The authors attempt to understand how self-efficacy affects the emotional and thinking patterns of employees. As a result, the positive responses that allow them to strive with their family and work requirements also eventually perform.

Semlali et al. (2016) examined women in Morocco's IT industry work and their work-life balance in conjunction with the positions they play in their professional and private lives. The authors stated that increase in the length of maternity leave, nurseries in the workplace, and telework can be used by various IT companies in the hope of enhancing the performance of female employees' work-life balance and welfare.

Thakur et al. (2015) examined the impact of perceived organizational support, aspects related to position and work involvement in work-life balance. The analysis indicated that employees with low efficacy may be looking for more in contrast to high self-efficacy, organizational support.

Sundaresan (2014) studied the components influencing work-life balance among working women and found the implications of poor balance of work and life. The author has framed a three-factor model as skills, knowledge, and behavior for both family and in work-life and also framed a role analysis model as working employee women's work-life imbalance is triggered by overload of responsibilities in both personal and work lives.

17.4.5 WORK-LIFE BALANCE

Wahyuni et al. (2022) measured the causes and effects of work-life balance for those employed in the garment industry. Working odd hours (beyond regular business hours) might throw workers' social and biological rhythms out of whack. The study's findings suggest that a work-life balance could improve employees' feelings of both job and life satisfaction.

Chowdhury et al. (2021) examined the factors that impact work-life balance with Bangladeshi professional women employees in the private sector. The study found that their employment has the greatest impact on their personal and family-social lives. These two determinants can have a considerable impact on total work-life balance.

Kaushal (2021) studied the numerous workplace pressures and how they relate to IT workers' ability to combine work and life. The research showed that job support is the most important component in explaining IT professionals' work-life balance.

Sharma (2021) investigated the elements that influence the quality of work life in organizations. The author listed some of the work and family life factors and found that there is no difference in employee perceptions between the public and private sectors.

Sharma (2021) examined the implications of the women's glass ceiling on women's employment advancement in the Indian service sector, as well as the moderating the effects of family structures (nuclear and joint) on the relationship between the glass ceiling and women's career advancement.

As mentioned above, personal, organizational, and societal impediments all fall under the category of glass ceiling women as a barrier and found that, personal obstacles did not have a substantial impact on the advancement of women managers in their careers compared to other two barriers.

17.5 RESEARCH METHODOLOGY

17.5.1 THEORETICAL BACKGROUND

Karassavidou et al. (2014) explained that border theory is a key; the emphasis is that boundaries and bridges between work and family need to establish and sustain a desired balance and be properly controlled. Border theory analyzes the organizational and personal variables that decide the complicated existence of work and family balance.

As per work-family border theory, the individual is an everyday boundary crosser between family and work. The theoretical model presented in this research is from Bharathi et al. (2016). And from that model, it understood that both personal and professional challenges have influence the work-life balance of IT women employees in the era of digital economy.

17.5.2 DATA COLLECTION PROCEDURE

In this research, descriptive research was conducted. The data are obtained from both the primary and secondary sources. The primary data were collected in the form of a questionnaire, and secondary data were arranged from journals, magazines, and websites (Bhambri et al., 2022).

The questionnaire consists of two parts. One part has the employee's demographic information. Another part has the forced-choice questions which relate to the independent variable of personal challenges of family involvement, family culture, and the professional challenges of long and irregular working hours, and job roles and responsibilities. These are questions with the dependent variable of work-life balance.

Questions in each section i.e. the questionnaire had prepared in the way of Google forms and it was sent to the IT women employees of three IT company which are located in Chennai, Tamil Nadu District, India and asked them to fill up then submitting back their answers. As a result, researchers have received about 112 questionnaire and after filtered out the questionnaire about 100, finally sample size is 100.

17.5.3 SAMPLE

For this research, data sample is simple random sampling that was employed, the sample size was 100 respondents and the area of the study was IT women employees of three IT companies in Chennai as mentioned above.

The analysis was conducted using SPSS_v (20.0) software, and the statistical tools adopted for this study were multiple correlation and multiple regression.

17.5.4 DATA STRUCTURE

The data structure of the questionnaire of this study contains two parts to measure the variables, the first half contains demographic information and the second portion contains various personal and professional challenges with work-life balance:

Section A – This section consists of the IT women employees' demographic information, such as

- Designation
- Department
- Age
- Experience in this Company
- Total Experience
- Education
- Marital Status
- Number of Children
- Number of Members in a Family
- Mode of Transport

Section B – This section consists of the forced-choice questions, which are related to the independent variable of personal and professional challenges and the dependent variable of work-life balance. The questions for each variable with the topic are listed in [Table 17.1](#) below.

17.5.5 MEASURES

Personal Challenges: Under the personal challenges, the factors of family involvement and family culture was measured. Respondents responded to several statements concerning their involvement with family members, spending enough time with them, whether the organization gives support to their family culture, and respect of colleagues regarding the family culture.

TABLE 17.1
The Questions for Each Variable with the Topic

Variables	Family Involvement	Family Culture	Long and Irregular Working Hours	Job Role and Responsibilities
1	Care provided by family members	Allows to work in IT industry	More working hours	Part of the team
2	Sufficient arrangements	Support to culture	Health issues	Target
3	Free hours	Respect with culture	Stress	Deadline
4	Enough time	Cope up with culture	Sleeping disorder	Quality of work
5	Support	Family gets together	Spent more time	Professionalism

Source: Author’s owner.
Table 17.1 depicts the questions for each variable with the topic.

The variables in this study were assessed using the five-point Likert-type scale, where 5 = strongly agree, 4 = agree, 3 = partially agree, 2 = disagree, and 1 = strongly disagree.

Professional Challenges: Under the professional challenges, the factors of long and irregular working hours and job role and responsibilities were measured. Based on these professional challenges, respondents responded to several statements concerning their organizations’ working hours, health issues when working long hours, completing a target within deadline, and concern about the quality of the work. The variables in this study were assessed using the five point Likert type scale, where 5 = strongly agree, 4 = agree, 3 = partially agree, 2 = disagree, and 1 = strongly disagree.

Work-Life Balance: Respondents responded to various statements concerning their ability to manage multiple roles on the job as well as at home, satisfaction with supervisors’ attitude, completing the work on time, and ability to maintain a good quality of family life. The variables in this study were assessed using the five-point Likert-type scale where 5 = highly satisfied, 4 = satisfied, 3 = partially satisfied, 2 = dissatisfied, and 1 = highly dissatisfied.

17.5.6 RESEARCH METHODOLOGY STRUCTURE

The structures of research methodology in [Figure 17.2](#) are as follows.

17.6 ANALYSIS AND INTERPRETATION

Based on the [Table 17.2](#), the reliability statistics were carried out with the help of Cronbach’s alpha. The reliability score is 0.948. Therefore, the questionnaire can be used for further statistical analysis.

Hypothesis 01: There is no significant relationship between the employees’ personal and professional challenges.

The significant P value is 0.000, which is lower than 0.05, therefore reject null hypothesis, and there is a significant relationship between family involvement and family culture with work-life balance, as shown in [Table 17.3](#).

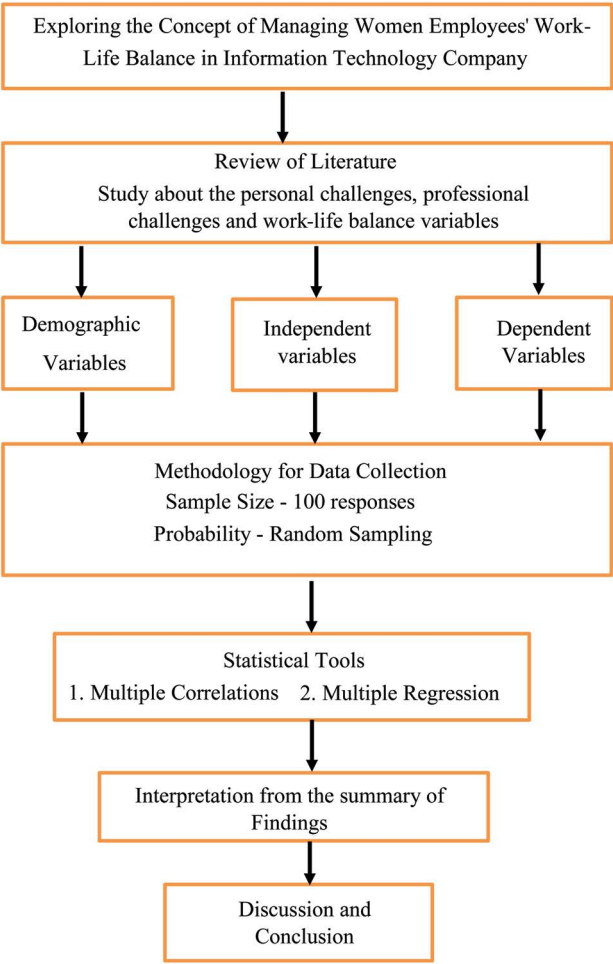


FIGURE 17.2 Research methodology structure for work-life balance of IT women employees.

Author’s ownership.

TABLE 17.2
Reliability Analysis

Cronbach's Alpha	Number of Items
0.948	67

TABLE 17.3

Multiple Correlations

		Family Involvement	Family Culture	Irregular and Long Working Hours	Job Role and Responsibilities
Family involvement	Pearson correlation	1	0.634**	0.634**	0.371**
	Sig. (2-tailed)	100	0.000	0.000	0.000
	N	100	100	100	100
Family culture		0.634**	1	164	0.407**
		0.000	100	0.102	000
		100		100	100
Irregular and long working hours		0.076	0.164	1	0.493**
		0.451	0.102	100	0.000
		100	100		100
Job role and responsibilities		0.371**	0.407**	0.493**	1
		0.000	000	0.000	100
		100	100	100	

Table 17.3 depicts the reliability statistics carried out with the help of Cronbach’s alpha.

The significant P value is 0.451, which is higher than 0.05, therefore accept null hypothesis, and there is no significant relationship between family involvement and irregular and long working hours. The significant P value is 0.000, which is lower than 0.05, therefore reject null hypothesis, and there is a significant relationship between family involvement and job role and responsibilities.

The significant P value is 0.102, which is higher than 0.05, therefore accept null hypothesis, and there is no significant relationship between family culture and long and irregular working hours. The significant P value is 0.000, which is lower than 0.05, therefore reject null hypothesis, and there is a significant relationship between family culture and job role and responsibilities.

The significant P value is 0.493, which is higher than 0.05, therefore accept null hypothesis, and there is no significant relationship between long and irregular working hours and job role and responsibilities.

Regression

Dependent variable: WLB

Predictors: (Constant), Family involvement, family culture, long and irregular working hours, and job role and responsibilities.

In Table 17.4, R denotes the simple correlation, and it is 0.517, which represents a moderate degree of correlation. R-Squared is the degree of determination, and it shows the extent to which examines the factors of family involvement, family culture, long and irregular working hours, and job role and responsibilities are significantly influencing work-life balance, as shown in Table 17.5.

Therefore, work-life balance is determined to an extent of 51.7% by the variables of family involvement, family culture, long and irregular working hours, and job role and responsibilities, as shown in Table 17.6.